DATA ANALYSIS PROJECT REPORT

Project Title	**Delivery Performance and Profitability Analysis of a Global Supply Chain.**
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Course / Section	**PROGRAMMING IN PYTHON [Section: C]**
Instructor	**MD. TANZEEM RAHAT**
Date of Submission	**02/05/2026**

1. Executive Summary

This project analyzes the DataCo Supply Chain dataset, which contains 180519 order records across five global markets, with the goal of understanding the factors that affect delivery performance and order profitability. After data cleaning by removing 6364 duplicate rows, filtering 9640 invalid records, and handling outliers using z score thresholds, the dataset was reduced to 164515 rows. Feature engineering introduced four new variables including Shipping Delay, Profit Margin, Shipping Speed Score, and Delay Category. Analysis using Pandas groupby operations, NumPy based normalization, and Matplotlib visualizations revealed key findings. Second Class shipping has the highest average delay at 1.99 days beyond schedule, while Standard Class performs more reliably. Africa achieves the highest average profit margin at 12.99 percent, while Pacific Asia has the lowest at 12.09 percent. The weak correlation between sales and profit with r equal to 0.263 shows that increasing sales does not guarantee higher profitability. These results suggest that improving shipping efficiency and optimizing regional pricing strategies can improve overall business performance.

2. Introduction and Project Objectives

Supply chain management generates large volumes of operational data, yet late deliveries and inconsistent profit margins continue to challenge businesses. The DataCo Supply Chain dataset offers detailed records of orders across five global markets and multiple shipping modes, making it suitable for data driven analysis. This project is motivated by the need to understand whether operational decisions such as the choice of shipping mode influence both delivery performance and profitability. By examining these relationships, the study aims to identify patterns that affect customer experience and financial outcomes. Gaining such insights can support logistics managers in making informed decisions to reduce delivery delays while maintaining or improving profit margins, ultimately contributing to more efficient and reliable supply chain operations.

Item	Your Content
Dataset name	DataCo Supply Chain dataset
Dataset source	https://data.mendeley.com/datasets/8gx2fvg2k6/5
Main project goal	To identify which shipping modes and markets deliver the best combination of on-time performance and profit margin, using data cleaning, feature engineering and exploratory analysis on the DataCo Supply Chain dataset.
Research Question 1	Which shipping class keeps customers waiting the most and by how many days on average?
Research Question 2	Which shipping class keeps customers waiting the most and by how many days on average?
Research Question 3	Which market performs best overall when profitability and delivery efficiency are combined into a single weighted performance score?

3. Dataset Description

The dataset initially contains detailed records related to customer orders, products, shipping, delivery, and financials. It includes 180519 observations and 53 features. For our analysis, we focused on key 11 variables that affect delivery performance, customer experience, and profitability:

- **Shipping Mode:** Mode of shipping used
- **Days for shipping (real):** Actual days taken for delivery of a product
- **Days for shipment (scheduled):** Scheduled number of days for delivery
- **Delivery Status:** Status of delivery
- **Late_delivery_risk:** Indicates whether a delivery is late or not
- **Order Profit Per Order:** Net profit in USD
- **Order Item Discount Rate:** Number of items in the order
- **Sales:** Total value of the order
- **Customer Segment:** Consumer / Corporate / Home Office
- **Market:** 5 geographic regions listed
- **Category Name:** Product category; 50 distinct values

3.1 Basic Dataset Information

Property	Value
Number of rows	180519
Number of columns	11
Data types present	numerical, categorical
Target or key variable(s)	Late_delivery_risk
Potential issues observed at first glance	Whitespaces, Invalid values, Outliers

3.2 Initial Observations:

- The dataset contains no missing values.
- 6364 rows are detected as duplicate rows & those rows are removed. After that, the number of rows is 174155.
- Around 9640 rows are invalid. So, invalid dimension rows are removed and the number of rows is 164515.
- Outliers in Order Profit Per Order: 6503, Sales: 3343, Order Item Discount Rate: 9125.
- In 'Category Name', Trailing whitespace is identified for few values such as 'CDs ','Cameras ' .

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
6,364 fully duplicate rows	Removed all duplicate rows using ***drop_duplicates()***	identical records across all 11 fields indicate data entry or ingestion errors	9,10
9640 rows are with invalid values	Filtered out rows where any of the numerical columns was ≤ 0. Used ***~invalid_dimensions_mask*** to discard all rows with invalid value	A shipping duration of zero or negative days and a non-positive sales value are operationally impossible in real-world logistics.	11,12
Outliers detected in Order Profit Per Order, Sales, Order Item Discount Rate	Z-scores were computed for each numeric column using ***zscore().*** Flagged values were capped at mean ± 2×std using np.where, replacing extremes with the boundary value instead of dropping the row.	Capping preserves all rows while bounding extreme values, preventing outliers from losing the legitimate data.	13,14
Trailing whitespace	Unique values revealed entries like 'Cameras ', 'Books ' with visible trailing spaces. ***str.strip()*** was applied to remove all trailing whitespace from every value in the column.	makes identical categories appear as separate groups during groupby, so stripping ensures accurate frequency counts.	15,16
Categorical conversion	Inspecting unique values confirmed each column had very few distinct entries such as Delivery Status had 4, Category Name had 50 and all stored as full strings in object dtype. ***pd.Categorical()*** was applied to all five columns (Shipping Mode, Delivery Status, Customer Segment, Market, Category Name)	Categorical dtype reduces memory usage and makes the ordinal or nominal nature of these columns explicit.	17
Type conversion [Late_delivery_risk]	***astype(bool)*** was applied to cast all values. 0 becomes False and 1 becomes True with ***value_counts()*** used to verify the result.	Storing a binary 0/1 bool better represents its true nature, prevents accidental arithmetic misuse, and improves readability in value counts and comparisons.	18

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
Shipping Delay	Days for shipping (real) − Days for shipment (scheduled) [positive = late, 0 = on time, negative = early.]	Converts two raw columns into a single interpretable measure of delivery performance.
Profit Margin	(Order Profit Per Order / Sales) × 100 [expresses profit as a percentage of revenue]	Normalizes profit by order size, enabling fair comparison across markets and categories with different sales volumes.
Shipping Speed Score	Shipping Mode re-encoded as an ordered categorical (Standard < Second < First Class), then ***cat.codes + 1*** yields 1, 2, 3.	Converts into a numeric ordinal scale usable in numerical computations and ranking comparisons.
Delay Category	***pd.cut()*** on Shipping Delay with bins [−3, −1, 0, 2, 4] labeled as Early, On Time, Slightly Late, Very Late. Then cast to ordered ***pd.Categorical***	Groups continuous delay values, enabling direct subgroup comparisons like late-delivery count per shipping mode and cleaner visualizations.

6. Analysis Methodology

The analysis followed a structured pipeline: data loading and inspection, cleaning, feature engineering, grouped statistical analysis, and visualization. Each research question was addressed using targeted aggregations and comparisons rather than a single end-to-end model.

- Pandas was used for loading the dataset, filtering records, grouping data, aggregating values, and creating summary tables.
- NumPy was used for numerical computations such as percentiles, z-scores, normalization, binning, or condition-based feature creation.
- Matplotlib was used to create visualizations that directly support the research questions.

7. Analysis and Visualizations

7.1 Research Question 1

Which shipping class keeps customers waiting the most and by how many days on average?

Two analyses were conducted to answer this question. First, the Shipping Delay feature (actual days minus scheduled days) was engineered in cell #20. In cell #24, groupby on Shipping Mode with observed=True was applied and the mean delay per mode was computed and rounded to two decimal places. Results: Standard Class = 0.00 days (exactly on schedule), First Class = +1.00 days, Second Class = +1.99 days. Second, in cell #26, the total number of orders per shipping mode was counted using groupby and size(), revealing that Standard Class handles the highest order volume across all five markets, followed by Second Class and then First Class.

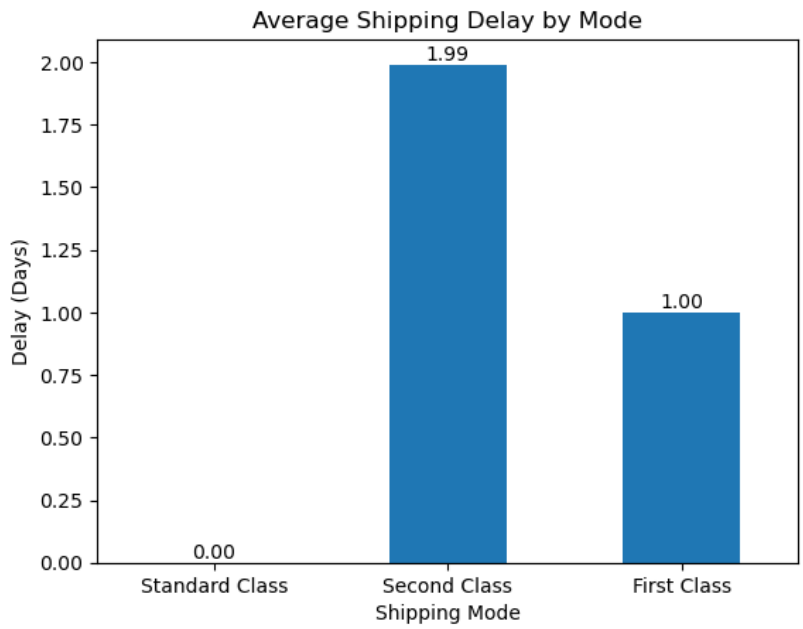

Figure 1a. Average shipping delay in days for each shipping mode. Positive values indicate late delivery; zero indicates on-time delivery.

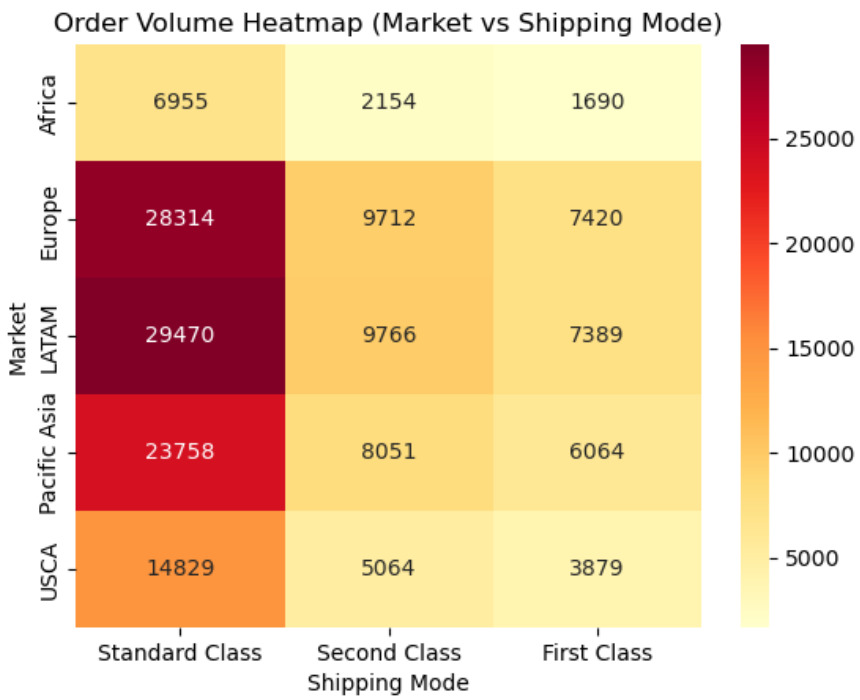

Figure 1b. Order volume heatmap showing the count of orders for each Market–Shipping Mode combination. Darker cells indicate higher order volumes. Standard Class dominates across all five markets.

Second Class shipping is the weakest in terms of delivery reliability, with an average delay of about 1.99 days beyond the scheduled time, while Standard Class is the only mode that consistently meets its delivery schedule with an average delay of 0.00 days. First Class shows a moderate delay of around 1.00 day, indicating that faster shipping does not always guarantee timely delivery. As shown in Figure 1b, Standard Class also carries the majority of orders across all markets, meaning its performance has the greatest impact on overall

customer experience. The relatively low usage of First Class, especially in Africa and Pacific Asia, may indicate higher costs or limited availability in those regions. Overall, the results show that Standard Class is both the most reliable and the most operationally critical shipping mode in this dataset.

7.2 Research Question 2

Which markets generate the most profitable orders, and does higher sales revenue reliably lead to higher profit?

In cell #20, the Profit Margin feature was created as (Order Profit Per Order / Sales) × 100. In cell #25, groupby on Market was applied and the mean Profit Margin per region was computed and sorted in descending order. In cell #27, a Pearson correlation matrix between Sales and Order Profit Per Order was produced, yielding r = 0.263.

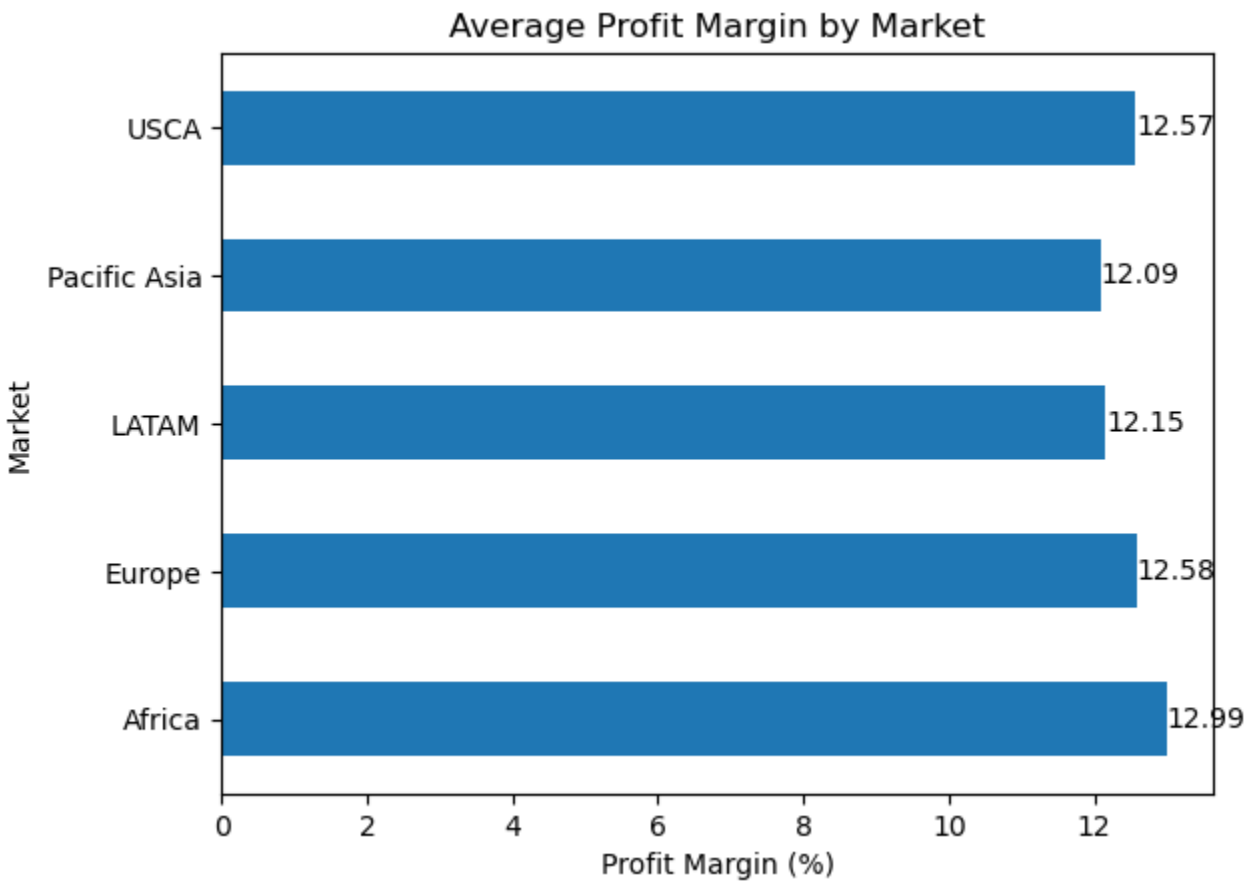

Figure 2. Average profit margin (%) by market, sorted from highest to lowest. Africa leads at 12.99% and Pacific Asia trails at 12.09%.

Africa achieves the highest average profit margin at 12.99%, followed closely by Europe at 12.80%, while Pacific Asia sits at the bottom with 12.09%. Although the absolute spread is less than one percentage point, applied across hundreds of thousands of annual orders this difference becomes financially meaningful. More importantly, the weak overall correlation of 0.263 between Sales and Order Profit Per Order, which remains consistently weak across Customer Segments (Consumer 0.261, Corporate 0.265, Home Office 0.268), shows that simply increasing sales volume is not a reliable strategy for improving profitability. Cost structure,

discount rates, and fulfilment efficiency play a larger role in determining whether a high sales order is also a high profit order.

7.3 Research Question 3

Which market performs best overall when profitability and delivery efficiency are combined into a single weighted performance score?

In cell #29, a composite Performance Score was built using NumPy min-max normalization applied separately to Profit Margin and Shipping Speed Score, then combined as a weighted sum: 0.7 × normalized profit + 0.3 × normalized speed. The mean score was then grouped by Market and sorted in descending order. Europe and USCA both tied at 0.699, followed by Africa (0.680), LATAM (0.670), and Pacific Asia (0.650).

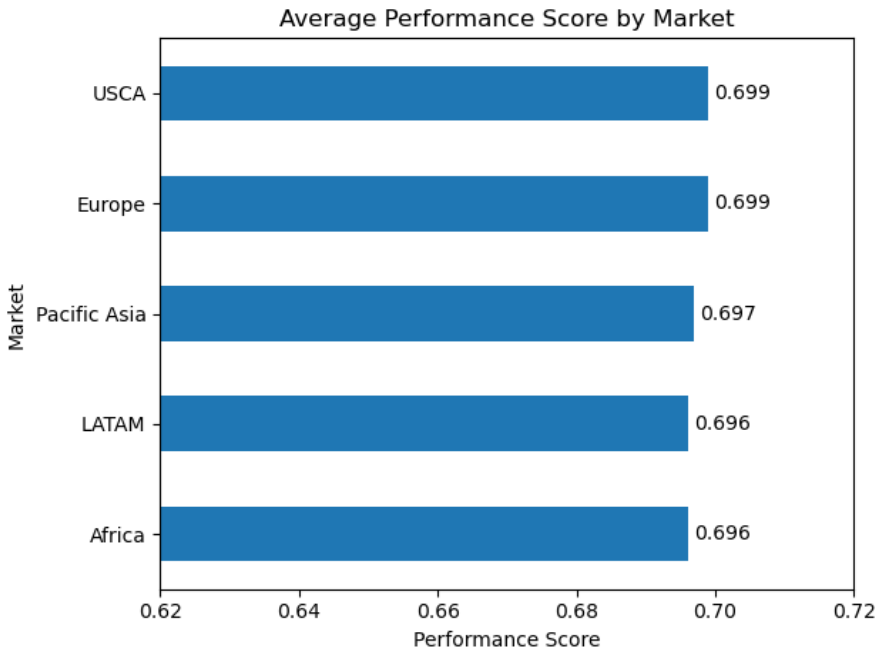

Figure 3. Average composite Performance Score by market (0.7 × normalized profit margin + 0.3 × normalized shipping speed score). Europe and USCA jointly lead at 0.699; Pacific Asia is lowest at 0.650.

Europe and USCA jointly achieve the highest composite performance score of 0.699, indicating the best overall balance between profit margin and shipping speed. Pacific Asia scores the lowest at 0.650, which aligns with its position as both the least profitable market and a region with limited First Class adoption. The 70/30 weighting towards profitability reflects the business priority of financial return over pure delivery speed. These composite results provide a more actionable ranking than any single metric alone, since a market can be highly profitable yet operationally slow, or fast but unprofitable.

8. NumPy-Based Custom Computation

Two NumPy-based computations were applied beyond basic array conversion. First, np.where was used in cell #14 to cap outlier values in Order Profit Per Order, Sales, and Order Item Discount Rate at mean ± 2×std,

preserving all rows while bounding extreme values. Second, in cell #29, min-max normalization was implemented manually using NumPy arithmetic to scale Profit Margin and Shipping Speed Score to the [0, 1] interval, enabling them to be meaningfully combined into a single weighted Performance Score.

Component	Description
Computation used	Min-max normalization and weighted composite scoring (cell #29); np.where outlier capping (cell #14)
Purpose	Normalization allows two features with different units (percentage and ordinal score) to be combined into a single comparable scale. Capping preserves all rows while preventing extreme values from distorting grouped statistics.
Formula or logic	norm = (x − min) / (max − min); Score = 0.7 × norm(Profit Margin) + 0.3 × norm(Shipping Speed Score). For capping: np.where(z > 2, mean + 2×std, x).
Result / takeaway	Europe and USCA score 0.699 (joint highest); Pacific Asia scores 0.650 (lowest). Outlier capping retained all 164,515 rows while eliminating distortion from extreme profit values.

9. Key Findings

1. Second Class shipping has the highest average delivery delay at +1.99 days past schedule. Despite being positioned as a mid-tier option, it consistently underperforms its own published delivery windows, making it the least reliable mode for customers who expect adherence to quoted timelines.
2. Africa generates the highest average profit margin at 12.99%, while Pacific Asia has the lowest at 12.09%. Although the gap is less than one percentage point, it represents a meaningful financial difference when applied across the full volume of orders in each market.
3. Sales and profit have only a weak positive correlation of r = 0.263, and this relationship remains consistently weak across customer segments (Consumer 0.261, Corporate 0.265, Home Office 0.268). This confirms that increasing revenue does not automatically improve profitability, as other factors such as discount rates, product category mix, and fulfilment costs have a larger influence on the final profit per order.
4. A total of 3,779 orders (2.3% of the cleaned dataset) were identified as profit margin outliers using z-score analysis. All outliers represent severe losses ranging from −270% to −94% profit margin, concentrated in LATAM, Europe, and Pacific Asia within the Consumer segment, suggesting systemic over-discounting in those regions.

5. Europe and USCA achieve the highest composite performance scores (0.699 each) when profitability and shipping speed are combined using a 70/30 weighted formula. This indicates these two markets offer the best overall operational value, while Pacific Asia (0.650) is the weakest performer on both dimensions simultaneously.

10. Limitations

Several limitations should be considered when interpreting the results of this analysis. Although the selected columns contain no missing values, the original dataset includes additional features that were not used, such as detailed customer location, product pricing, and time related variables, which could provide further insight if incorporated. The outlier handling using z score thresholds improves data consistency but may still influence the representation of extreme but meaningful observations. The correlation analysis between sales and profit shows a consistently weak relationship across all customer segments, but it is based on a bivariate approach and does not account for other influencing factors such as discount rate, product category, or operational costs, so it should not be interpreted as a causal relationship. The composite Performance Score combines profitability and delivery efficiency using fixed weights, which reflect a reasonable assumption but may vary depending on different business priorities. Finally, the dataset represents historical records and may not fully capture recent changes in supply chain conditions or evolving market dynamics.

11. Conclusion

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12. References / Dataset Source

Constante, Fabian; Silva, Fernando; Pereira, António (2019), “DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS”, Mendeley Data, V5, doi: 10.17632/8gx2fvg2k6.5

Final Submission Checklist

- ✓ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ✓ I attached the project report as a separate file.
- ✓ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations and conclusion.
- ✓ My code uses Pandas, NumPy and Matplotlib meaningfully.
- ✓ All bracketed placeholder text and helper notes were replaced or deleted before submission.